

# Factoring in individual stressor exposure is of the essence

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ARISING FROM D. Wolke et al. *Nature Mental Health* <https://doi.org/10.1038/s44220-025-00479-3> (2025)

Wolke et al. have provided an excellent overview and analysis of the current conceptualizations and statistical methods in resilience research<sup>1</sup>. The paper gives much needed guidance and contains valuable recommendations for future research. However, we believe that the authors have overlooked one key methodological issue, which is that studies with only coarse operationalization and measurement of individual stressor exposure are likely to yield false classifications of individuals' resilience status and wrong estimates of the statistical effects of resilience factors. Existing data collection and analysis methods allow for addressing such confounding by under-specified exposure.

Wolke et al. make a strong point that any resilience study, regardless of which definition of resilience a researcher adopts, needs to involve adversity (understood as inherently negative experiences) as one core element, and they recommend to 'clearly operationalize adversity by specifying the type and severity of adverse experience (individual versus cumulative), developmental timing, duration (for example, early life versus chronic) and the measurement approach (for example, baseline versus repeated assessments)'. The authors argue that clear operationalization is important 'because [above] factors substantially influence observed resilience prevalence'. This is clearly a very strong argument, and we very much applaud the authors' emphasis on including well-specified adversity measures in any resilience study.

However, the observation that factors such as type, severity or duration of adversity influence resilience prevalence comes with a hidden twist. If cumulative, more severe, and longer exposure to stressors is associated with a smaller fraction of individuals classified as resilient in a person-centered approach, this raises the thorny question: could it be that, in a study with no or only coarse operationalization and measurement of individual exposure, individuals assigned to a resilient class (for instance, individuals showing a trajectory of consistently low or of quickly recovering mental health problems after a potentially traumatic event, PTE) are simply individuals that are less exposed than those assigned to non-resilient classes (for example, showing trajectories of sustained high or of increasing problems)?

Put more bluntly: as long as I do not have a detailed picture of the stressor exposure a participant experiences throughout a study's observation period, can it be that my resilience classification is false? This question is relevant because many studies only record the index exposure (for example, a PTE, the onset of a chronic disease, a disaster, a pandemic) with reference to which the outcome is established (for example, using a single post-exposure mental health measurement or repeated post- and sometimes also pre-exposure mental health measurements); they do not usually, however, consider that (1) there will necessarily be individual differences in the quantity and quality of that index exposure and (2) there will almost certainly be other exposures occurring in the 'background'<sup>2</sup> during the time of observation. (Life is stressful.) Next to major stressors, background stressors may include less severe stressors (microstressors, daily hassles), which can also have a profound impact on mental health<sup>3</sup>. For examples of person-centered studies that are under-specified on the exposure side, see the vast corpus of COVID-19 studies reviewed in ref. 4. The question of potentially false resilience classification is also relevant because, if under-specified person-centered approaches are used to identify factors that predict resilient class membership, false classification will inevitably lead to over- or underestimation of the statistical effects of these resilience factors.

The impact that individual differences in exposure to adversity can have on classification is illustrated by a study in responders to the 9/11 terror attack (index exposure)<sup>5</sup>. In that study, the authors differentiated between ten different potential attack-related PTEs and observed that exposure to a smaller number of these events made it more likely that a participant would show a stable trajectory of no or low post-traumatic stress symptoms over the next 12 years. Furthermore, this likelihood was increased in participants reporting fewer PTEs before 9/11 and fewer life stressors and PTEs in the 12 years after (background exposures). Another study conducted among Swedish survivors of the 2004 tsunami in Southeast Asia found that survivors with less direct disaster exposure (index exposure) and fewer post-disaster PTEs (background

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stressors) were also more likely to show stably low post-traumatic symptom trajectories over six years<sup>6</sup>. In a population-representative study among Hong Kong Chinese, trajectories of low internalizing symptoms over the course of the COVID-19 pandemic (index exposure) were associated with fewer financial difficulties in the later phase of the pandemic (background exposure)<sup>7</sup>. Taking up these findings, a study in Germany found that shorter-term trajectories of internalizing symptoms to major negative life events in adults and assignment of participants to the low-symptom class differed depending on whether trajectory modeling was based on raw symptom scores or on symptom scores corrected for concurrent reports of background stressors (other life events as well as daily hassles)<sup>8</sup>.

Problems stemming from under-specified stressor exposure are not limited to person-centered approaches. For instance, using a variable-centered approach, Lowe et al.<sup>2</sup> recently observed that male healthcare workers showed better mental health outcomes during COVID-19 than female healthcare workers, suggesting that male gender can be a promotive resilience factor. However, when the authors incorporated the higher levels of exposure reported by the female participants in their regression-based outcome prediction analysis, the significant gender effect disappeared. The male participants were not tougher, they had just been less exposed (for example, they had to juggle less between work- and family-related tasks).

These observations highlight a massive challenge that resilience research is confronted with: the less we know about the adversity encountered by participants during a study observation period, the less we can exclude that what appears to be resilience is in fact less exposure and the less certain we can be that factors found to associate with ‘resilience’ are truly resilience factors. The problem is particularly pertinent in studies on resilience to early-life adversity where outcomes are often determined years after the index exposure (for example, in adulthood).

These considerations have clear consequences for designing future resilience studies and for analyzing existing datasets. In any new data collection, emphasis should be placed on not merely specifying the type, severity, timing or duration of adverse experience or merely reporting whether exposure was assessed at baseline or repeatedly, as Wolke et al. rightfully suggest; rather, the concept of ‘adverse experience’ should be abandoned for the much more realistic concept of ‘adverse experiences’, which can only be captured through repeated stressor monitoring—the more frequent and the more exhaustive the better<sup>3</sup>. In the exploitation of existing datasets, attention should be given to those longitudinal studies that have already implemented extensive stressor monitoring. Examples are the Generation R and ALSPAC birth-cohort studies<sup>9</sup>, the ABCD study in adolescents<sup>10</sup>, the THRIVE (<https://www.universiteitleiden.nl/resiliencecenter/onderzoek/thrive>), MARP<sup>11</sup> and DynaM-OBS<sup>12</sup> studies in young adults, the SNF Resilience<sup>13</sup> and HMZ Stress<sup>14</sup> studies in medical staff, and the HBS<sup>15</sup> and LORA<sup>16</sup> studies conducted in broader adult populations. More recently, the frequent monitoring paradigm has also been extended to interventional studies (for example, ref. 11). Methods to incorporate repeated exposure measures into person-centered as well as variable-centered analyses are now available<sup>3,8,11,17</sup> and raise realistic hopes that resilience research can escape confounding by under-specified exposure.

We feel that courageously addressing the problem of confounding by under-specified exposure is perhaps the most important challenge for future resilience research and should therefore be the most important call and primary recommendation to be made by a conceptual review paper that has the ambition to guide future work.

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## Competing interests

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